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Engineering Mathematics

2Week [11 September - 17 September]

Quiz 2

Started on	2023-09-17 23:49
State	Finished
Completed on	2023-09-17 23:51
Time taken	1 min 26 secs
Grade	10.00 out of 10.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

Among the equations below choose a linear differential equation.

Select one:

- ☒ $x^2 \sin x - (\cos x)y = (\sin x) \frac{dy}{dx}$ ✓
- ☐ $y \frac{dy}{dx} = e^x + 1$
- ☐ $\frac{dy}{dx} + y^3 = 5x - 1$
- ☐ $\frac{dy}{dx} = \frac{x}{y}$

Your answer is correct.

The correct answer is: $x^2 \sin x - (\cos x)y = (\sin x) \frac{dy}{dx}$

Question 2

Correct

Mark 1.00 out of 1.00

Find the integrating factor of the equation

$$\frac{1}{x} \cdot \frac{dy}{dx} - \frac{2y}{x^2} = x \cos x, \quad x > 0.$$

Select one:

- ☒ x^{-2} ✓
- ☐ x^2
- ☐ $x \cos x$
- ☐ $e^{2/x}$

Your answer is correct.

The correct answer is: x^{-2}

Question 3

Correct

Mark 1.00 out of 1.00

Solve the initial value problem $\frac{dy}{dx} + \frac{3y}{x} = 3x, \quad y(1) = \frac{3}{5}.$

Select one:

- ☒ $y = \frac{3}{5}x^2$ ✓
- ☐ $y = x^2 - \frac{2}{5}$
- ☐ $y = \frac{2}{5}x^2$
- ☐ $y = \frac{8}{5} - x^2$

Your answer is correct.

The correct answer is: $y = \frac{3}{5}x^2$

Question 4

Correct

Mark 2.00 out of 2.00

The temperature T (in units of $100^\circ F$) of a university classroom on a cold winter day varies with time t (in hours) as $\frac{dT}{dt} = 1 - T$ when heating is ON. Suppose $T = 0$ at 9 A.M., the heating unit is ON between 9:00 and 12:00. How warm will the classroom be at noon?

Select one:

- ☒ $(1 - e^{-3})100^\circ F$ ✓
- ☐ $(1 - e^3)100^\circ F$
- ☐ $(e^{-3} - 1)100^\circ F$
- ☐ $(e^3 - 1)100^\circ F$

Your answer is correct.

The correct answer is: $(1 - e^{-3})100^\circ F$

Question 5

Correct

Mark 1.00 out of 1.00

The equation $M(x, y)dx + N(x, y)dy = 0$ is exact if and only if

Select one:

- ☒ $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ ✓
- ☐ $\frac{\partial N}{\partial y} = \frac{\partial M}{\partial x}$
- ☐ $\frac{\partial^2 M}{\partial y \partial x} = \frac{\partial^2 N}{\partial x \partial y}$
- ☐ $\frac{\partial^2 M}{\partial x \partial y} = \frac{\partial^2 N}{\partial y \partial x}$

Your answer is correct.

The correct answer is: $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Question 6

Correct

Mark 1.00 out of 1.00

Among the equations below choose an exact differential equation.

Select one:

- ☒ $(1 + e^x y + x e^x y)dx + (x e^x + 2)dy = 0$ ✓
- ☐ $(x + \sin y)dx + (y + \cos x)dy = 0$
- ☐ $(x^{10} - 2y)dx + xdy = 0$
- ☐ $(x^2 - 3y^3)dx + 2xydy = 0$

Your answer is correct.

The correct answer is:

$$(1 + e^x y + x e^x y)dx + (x e^x + 2)dy = 0$$

Question 7

Correct

Mark 1.00 out of 1.00

The equation $(x + 3x^3 \sin y)dx + (x^4 \cos y)dy = 0$ is not exact but multiplying this equation by some factor yields an exact equation. Choose the correct factor from below.

Select one:

- ☒ x^{-1} ✓
- ☐ x^{-2}
- ☐ x^2
- ☐ x

Your answer is correct.

The correct answer is: x^{-1}

Question **8**

Correct

Mark 2.00 out
of 2.00

Solve the initial value problem

$$(e^t x + 1)dt + (e^t - 1)dx = 0, \quad x(0) = 1.$$

Select one:

- ☒ $t + (e^t - 1)x = 0$ ✓
- ☐ $e^t x + 1 - 2x = 0$
- ☐ $e^t x - e^t = 0$
- ☐ $e^t + (e^t - 2)x = 0$

Your answer is correct.

The correct answer is: $t + (e^t - 1)x = 0$